

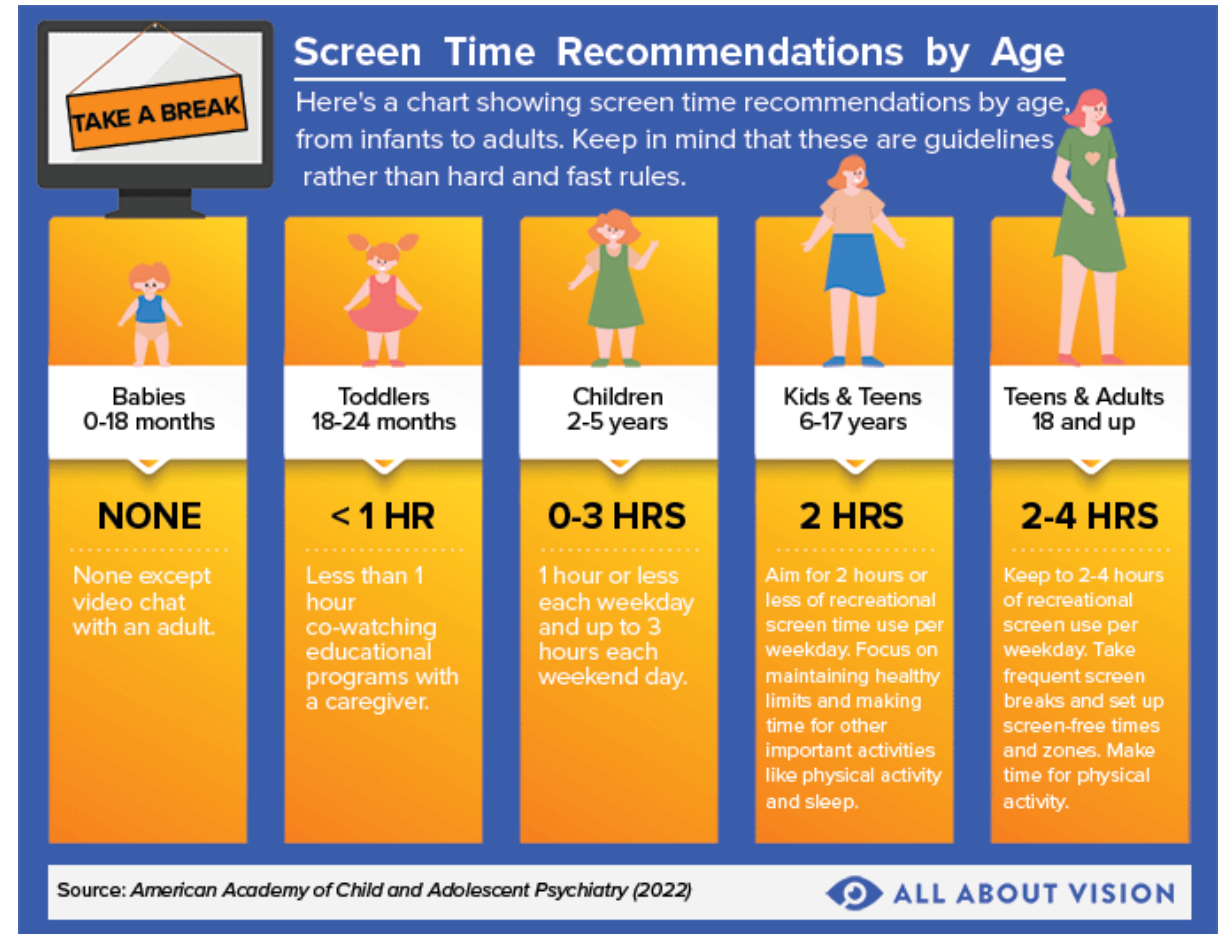
Data Sonification

*The art of turning numbers (data) into sound through
program/code*

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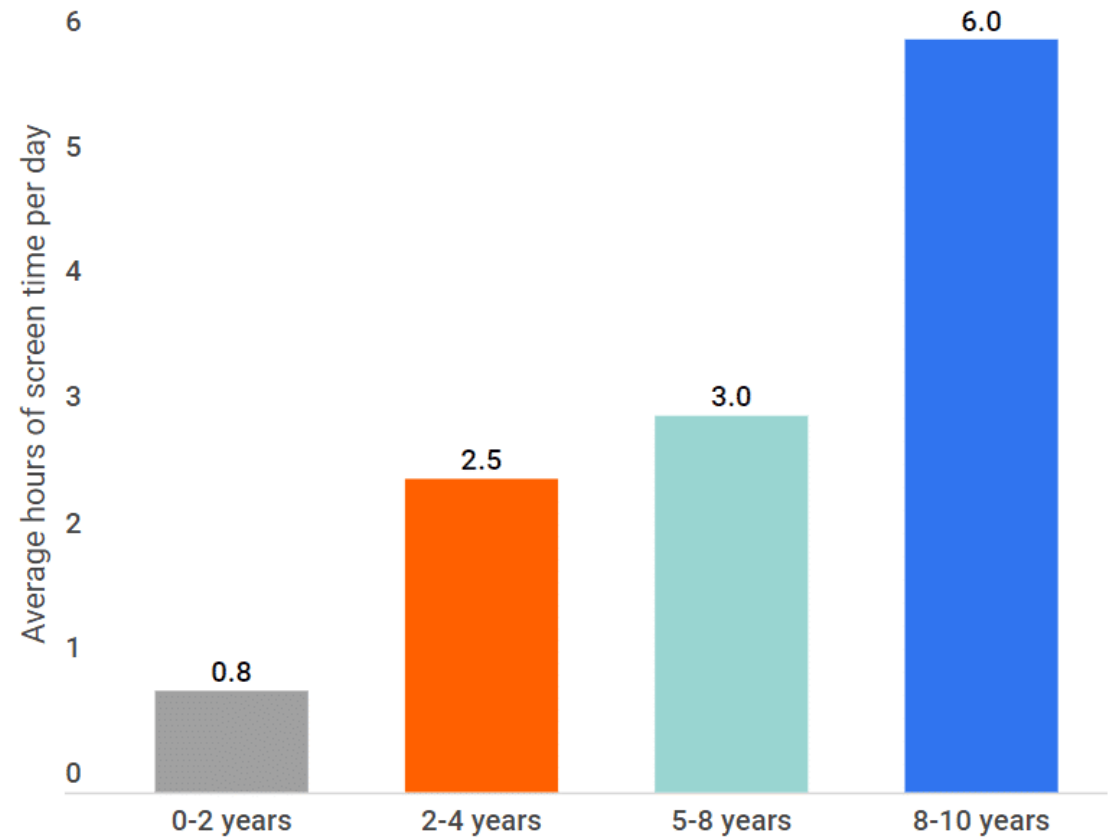
Can you give some examples for
Data?

How is data
usually
represented?



Visuals

AVERAGE CHILD SCREEN TIME BY AGE RANGE



Audio?

Was there any data?



What are we trying to do?

- We are going to write a code that takes **a list of numbers** and **turns each number into a sound pitch**.
- Example:
 - how many people boarded in each bus stop or
 - how many empty seats at each bus stop
- Each number is played for a specific duration of time (eg: *half a second*, one after another, so we can *hear* the data.
 - Small numbers → low sounds
 - Big numbers → high sounds

Who can be benefitted from this data sonification?

- Any thoughts?

Who can be benefitted from this data sonification?

- Helping visually impaired users
- Making art with data
 - Data ---> Music

Libraries- Tools in ToolBoxes

Bela.h → lets us work with real-time audio on the Bela board

cmath → gives us maths functions like `sin()`

vector → lets us store a list of numbers that can change size



Vector: List

- `std::vector<float> data = {2, 5, 7, 3, 9, 4, 6};`
- `data = {2, 5, 7, 3, 9, 4, 6} → This is our dataset.`
- Each number could represent “***how many people boarded in the first 7 bus stops in the route from High Holborn to Peckham in the 68 bus route to West Norwood***”
- We will play one sound for each number

X	Camberwell Road / Albany Road	☆	>
M	Westmoreland Road	☆	>
H	East Street	☆	>
C	Larcom Street	☆	>
Z	Heygate Street	☆	>
V	Elephant & Castle Station	☆	>
A	Elephant & Castle / London Road	☆	>
P	St George's Circus	☆	>
Q	The Old Vic	☆	>
K	Waterloo Station / Tenison Way 	☆	>
P	Waterloo Bridge / South Bank	☆	>
T	Lancaster Place	☆	>
F	Aldwych / Kingsway	☆	>
N	Holborn Station	☆	>

Variables for sound and timing

- What pitch or how high or low (frequency) for each data point?
- Which data point (current index) are we in now?
- How long to play music for each data point (samples per event)?
- How do we know we have reached end of the data points (eg: 7th bus stop- counter)?

```
data = {2, 5, 7, 3, 9, 4, 6};
```



Variables for sound and timing

- What pitch or how high or low (frequency) for each data point?
 - `float frequency = 220.0;`
- Which data point (current index) are we in now?
 - `int currentIndex = 0;`
- How long to play music for each data point (samples per event)?
 - `int samplesPerEvent;`
- How do we know we have reached end of the data points (eg: 7th bus stop-counter)?
 - `int counter = 0;`

```
data = {2, 5, 7, 3, 9, 4, 6};
```



Set up function

```
bool setup(BelaContext *context, void *userData) {  
    samplesPerEvent = context->audioSampleRate * 0.5;  
    return true; }
```

- This runs just once
- audioSampleRate is how many sound samples happen per second
- We multiply by 0.5 so each data point lasts half a second

Render function

`data = {2, 5, 7, 3, 9, 4, 6};`

```
for(unsigned int n = 0; n < context->audioFrames; n++) {
```

`// This runs again and again. This loop runs once for each audio sample. audio is built one tiny piece at a time`

```
if(counter++ >= samplesPerEvent) { counter = 0; currentIndex =  
(currentIndex + 1) % data.size();
```

`// When half a second has passed: Reset the counter and move to the next number in the data; % data.size() makes sure we loop back to the start when we reach the end`

Turning numbers to pitch

`data = {2, 5, 7, 3, 9, 4, 6};`

```
float value = data[currentIndex];  
frequency = 200.0 + value * 60.0;
```

- This is the most important idea:
 - We take a number from the data
 - `// data = {2, 5, 7, 3, 9, 4, 6}`
 - We convert it into a sound frequency
 - Example:
 - Value = 2 -> low pitch
 - Value = 9 -> much higher pitch
 - Bigger the number -> Higher the sound

Making the sound wave- Sine

```
float out = sinf(phase * 2.0f * M_PI);
```

- sin() creates a smooth sine wave
- Sine waves sound clean and simple

```
phase += frequency / context->audioSampleRate;
```

```
if(phase >= 1.0)
```

```
    phase -= 1.0;
```

- This moves the wave forward at the right speed
- Higher frequency -> wave moves faster -> higher pitch

Send sound to speaker

```
for(unsigned int ch = 0; ch < context->audioOutChannels; ch++) {  
    audioWrite(context, n, ch, out * 0.2); }
```

- Sends the sound to **all speakers**

Bela- Microcontroller with Audio capabilities



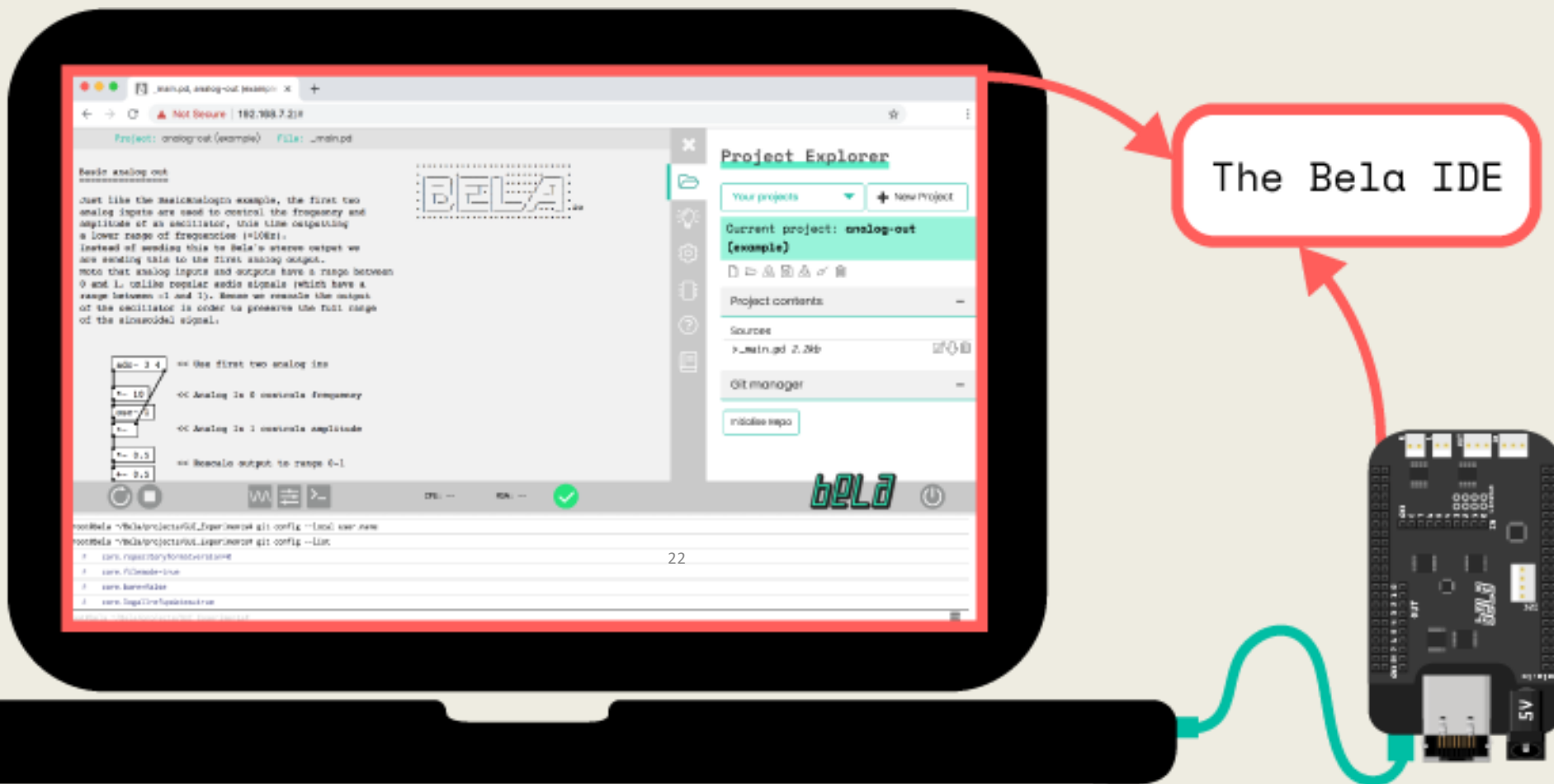


Slide Acknowledgement

The following slides are provided by
Dr. Teo Dannemann

THANK YOU!!!

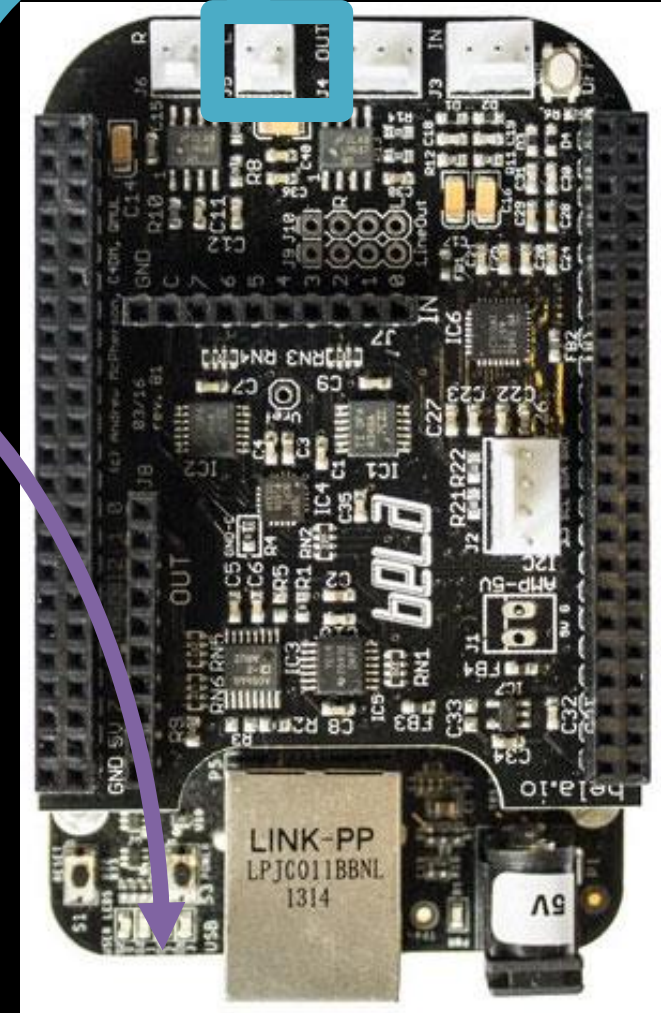




Let's start!

1. Connect your Bela unit to your computer using the appropriate USB cable and wait for it to boot up. When BELA is ready the blue LEDs will blink in a heartbeat pattern.

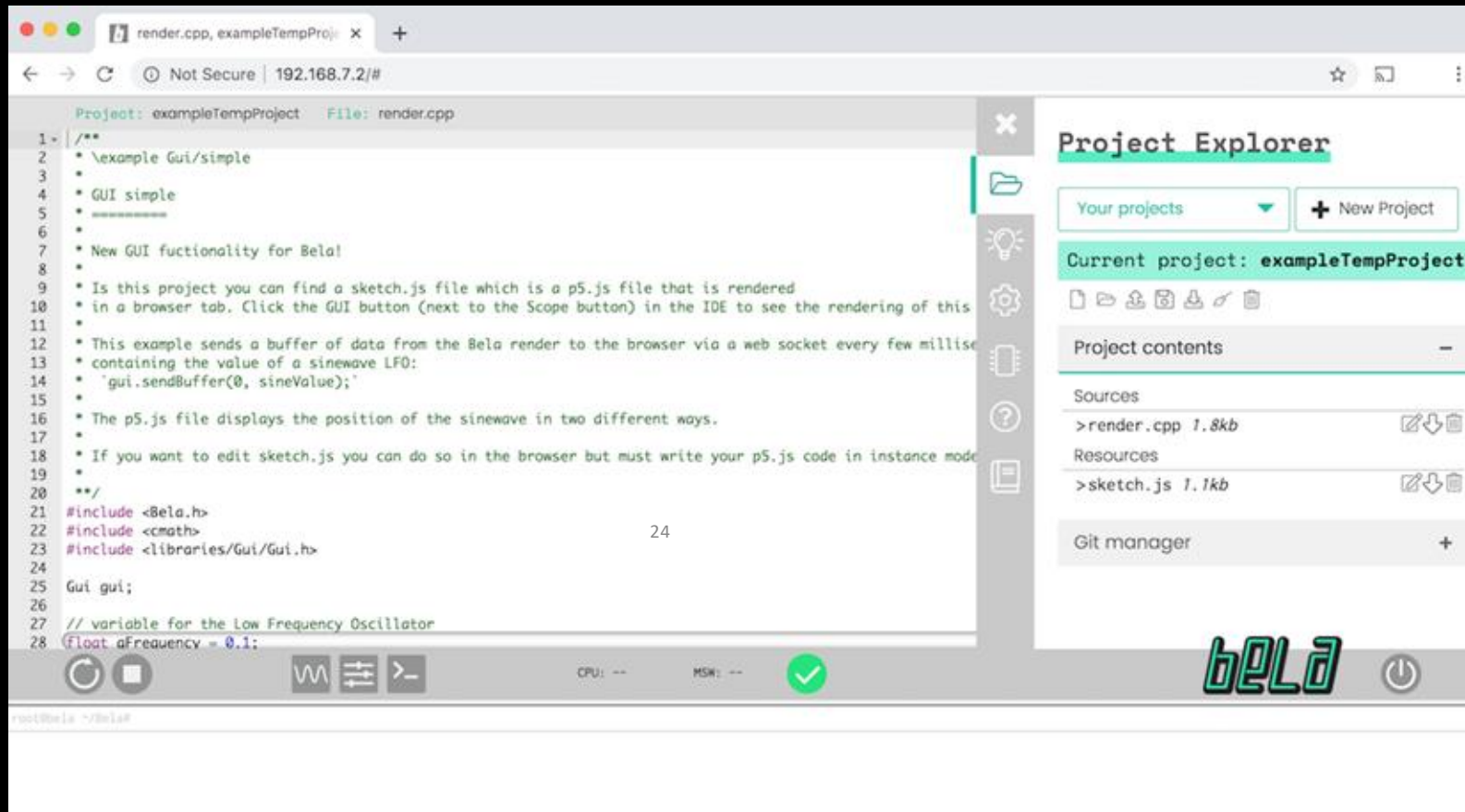
2. Connect speaker to the speaker OUT using the two jumper cables



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3. Go to <http://bela.local>

You should see something like this:



Let's explore some examples:

Using sensors:

1.Trill sensor

1.Bar

2.Hexa

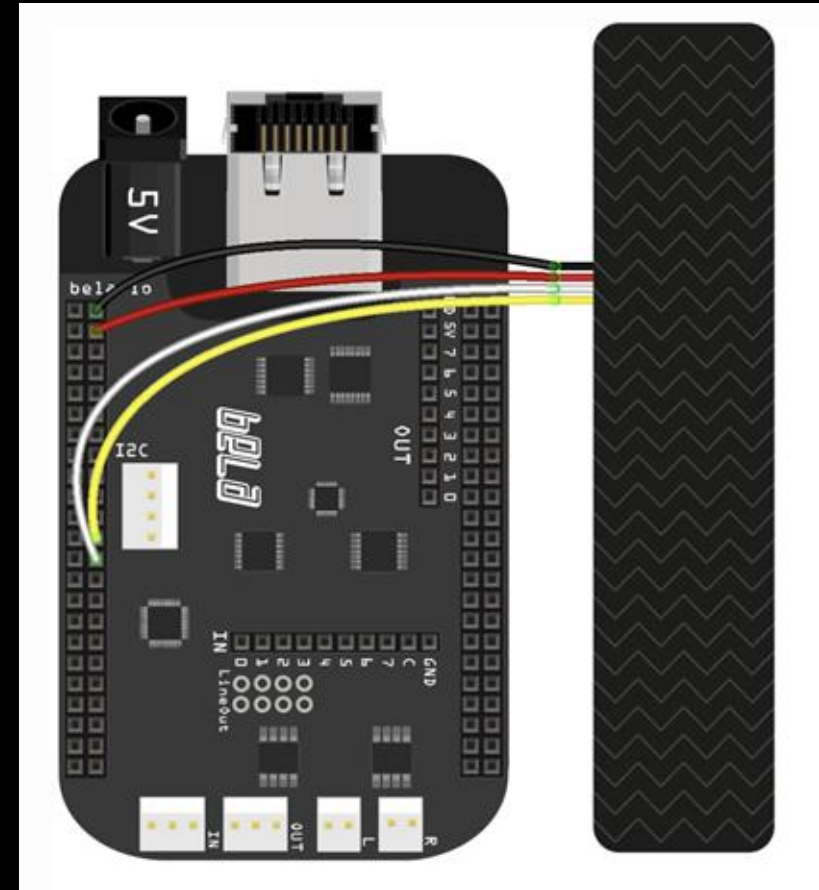
3.Square

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Trill sensor

1. Connect the cables following the diagram
2. In the IDE, go to Examples > Trill > bar-sound
3. Click the *Build & Run* button at the bottom left

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Activity time!

Goal: The goal is to collect some data that you will then input into Bela to make sound from it.

- 5 minutes discussion of ideas in groups
- 10 minutes to go around this room/3rd floor and collect (or record) data
 - Inspirations:
 - Tell them about three or four songs and record the most to least popular ones by count
 - Tell them about four or five travel destinations and record the most to least favorite by count
 - Observe the count of plants by type³⁰
- 10 minutes to develop your compositions
 - We will have pre-prepared code in Bela, so you don't have to know technicalities on music or sound, just focus on how to feed the data to the code.

Code Base and Audio Files

Access this link:

<https://github.com/kjayavel/Bela/>



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Bela website

<http://bela.local/>





Final points and close



Final points

Hope to see you soon!!!

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THANK YOU

